7 Text analysis

This chapter covers

- Overview of text analysis
- Anatomy of an analyzer
- Built-in analyzers
- Developing custom analyzers
- Understanding tokenizers
- Learning about character and token filters

Elasticsearch does a lot of ground (and grunt) work behind the scenes on incoming textual data. It preps data to make it efficiently stored and searchable. In a nutshell, Elasticsearch cleans text fields, breaks text data into individual tokens, and enriches the tokens before storing them in inverted indexes. When a search query is carried out, the query string is searched against the stored tokens, and any matches are retrieved and scored. This process of breaking the text into individual tokens and storing it in internal memory structures is called *text analysis*.

The aim of text analysis is not just to return search results quickly and efficiently, but also to retrieve *relevant* results. The work is carried out using *analyzers*: